Thermodynamic description of the Cu–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

Synthetic document generated by the ORCHID pipeline benchmark; not a real publication. CC BY 4.0. Derived from the SGTE binary dataset collection (B. Hallstedt, Calphad 89 (2025) 102833, CC BY 4.0) and the original assessment cited in the dataset.

Abstract

This synthetic document describes the Cu–Zn binary system as calculated with pycalphad from the SGTE dataset CuZn-08Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, (Cu) + L → β, lies at 899 °C.

1. Introduction

The Cu–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as peritectic, intermetallic rich, order disorder. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of W. Gierlotka, S.-W. Chen, J. Mater. Res., 23, 258-63(2008). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Cu, the liquidus meets the axis at 1085 °C. Pure Zn melts at 420 °C.

2. Phase diagram and invariant reactions

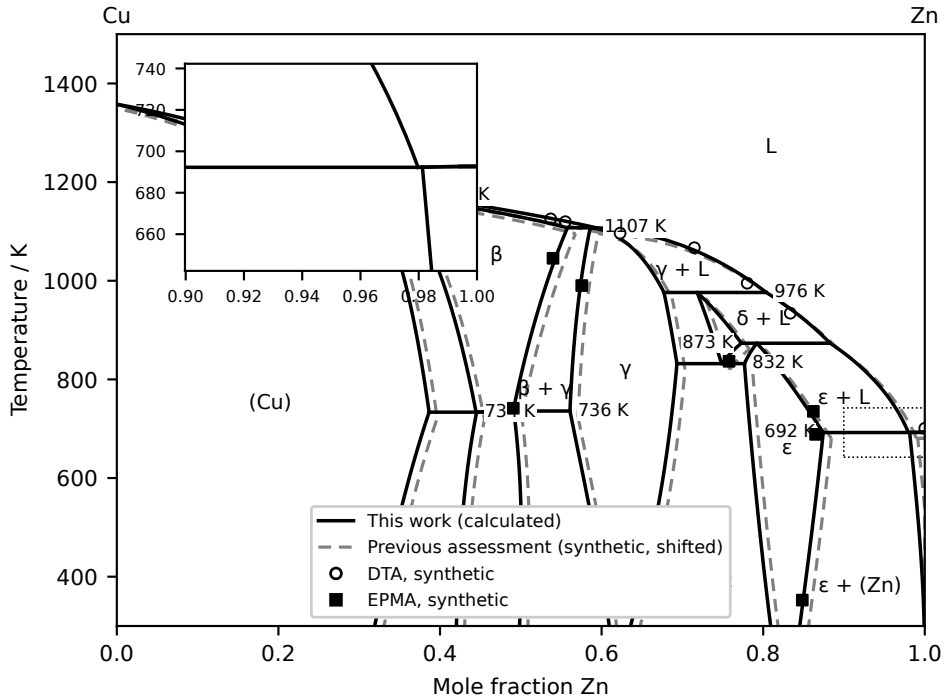


Fig. 1. Calculated Cu–Zn phase diagram with synthetic experimental-style points and a dashed, deliberately shifted second assessment (inset: enlarged view near the lowest invariant).

The peritectic reaction (Cu) + L → β is calculated at 899 °C, with L at x(Zn) = 0.375. We obtain 834 °C for the peritectic reaction (β + L → γ), where L has a mole fraction of 0.593. The dataset places a peritectic (γ + L → δ) at 703 °C; the composition of L is a mole fraction of 0.804. We obtain 600 °C for the peritectic reaction (δ + L → ε), where L has x(Zn) = 0.883. The dataset

places a eutectoid (δ → γ + ε) at 559 °C; the composition of δ is a mole fraction of 0.749. The dataset places a peritectoid (β + γ → β′) at 463 °C; the composition of β′ is a mole fraction of 0.491. Eutectoid equilibrium β → (Cu) + β′ is found at 460 °C and a mole fraction of 0.445 (β). The eutectic reaction L → ε + (Zn) is calculated at 419 °C, with L at x(Zn) = 0.980.

Table 1. Invariant reactions and congruent points, calculated (Cu–Zn).

Reaction	Type	T (°C)	x(Zn), 1	x(Zn), 2	x(Zn), 3
(Cu) + L → β	peritectic	899.4	0.319	0.375	0.354
β + L → γ	peritectic	834.2	0.558	0.593	0.586
γ + L → δ	peritectic	703.1	0.677	0.804	0.718
δ + L → ε	peritectic	600.2	0.773	0.883	0.792
δ → γ + ε	eutectoid	558.8	0.749	0.693	0.777
β + γ → β′	peritectoid	462.7	0.491	0.561	0.491
β → (Cu) + β′	eutectoid	460.4	0.445	0.387	0.445
L → ε + (Zn)	eutectic	419.1	0.980	0.874	0.981

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Cu–Zn).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Zn)	at T / °C
(Cu)	cF4	A1	Cu	< 27[a]	1084.6	0.000–0.386	467
β′	cP2	B2	CsCl	< 27[a]	462.7	0.421–0.499	57
β	cI2	A2	W	460.4	899.4	0.378–0.556	827
γ	cI52	$D8_2$	Cu5Zn8	< 27[a]	834.2	0.564–0.693	557
δ	cI2	A2	W	558.8	703.1	0.739–0.770	607
ε	hP2	A3	Mg	< 27[a]	600.2	0.784–0.874	417
(Zn)	hP2	A3	Mg	< 27[a]	419.5	0.981–1.000	417

[a] Stable at the lowest calculated temperature.

β′ is stable up to 463 °C. The solubility of Cu in (Zn) reaches a mole fraction of 0.019 at 419 °C.

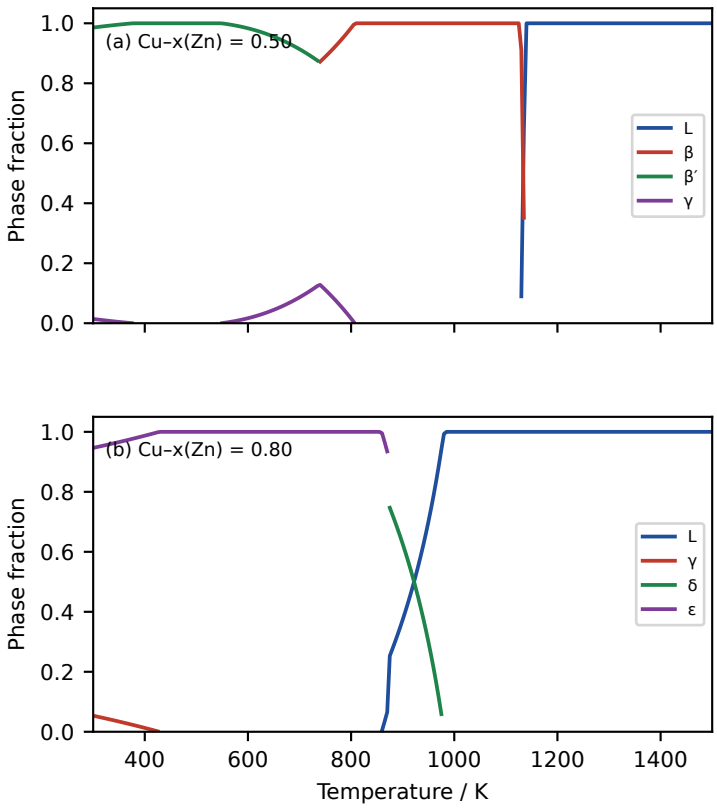


Fig. 2. Calculated equilibrium phase fractions versus temperature for Cu–x(Zn) = 0.50 and x(Zn) = 0.80.

4. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −6436 J/mol, at x(Zn) = 0.440. The calculated enthalpy of formation of δ at 298.15 K is −5.76 kJ/mol per mole of atoms. Its heat capacity at room temperature is 25.16 J/(mol K). For the equiatomic liquid at 1450 K, the Gibbs energy of mixing is −14.55 kJ/mol-atom.

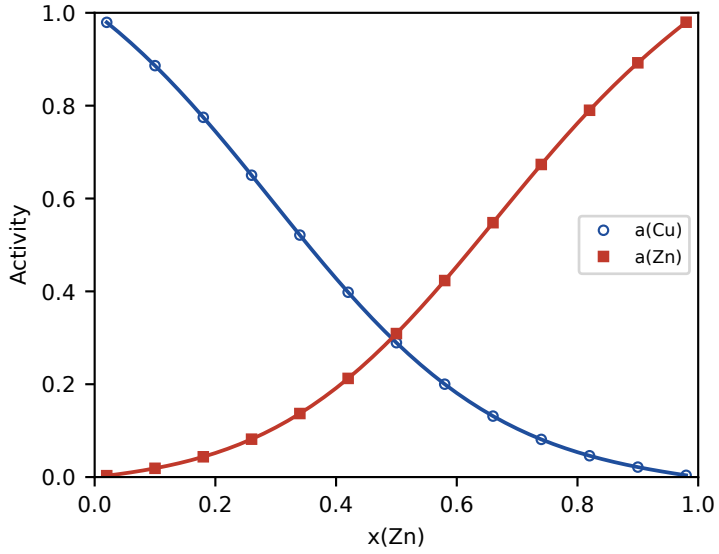


Fig. 3. Calculated activities in the liquid at 1450 K (reference: pure liquids).

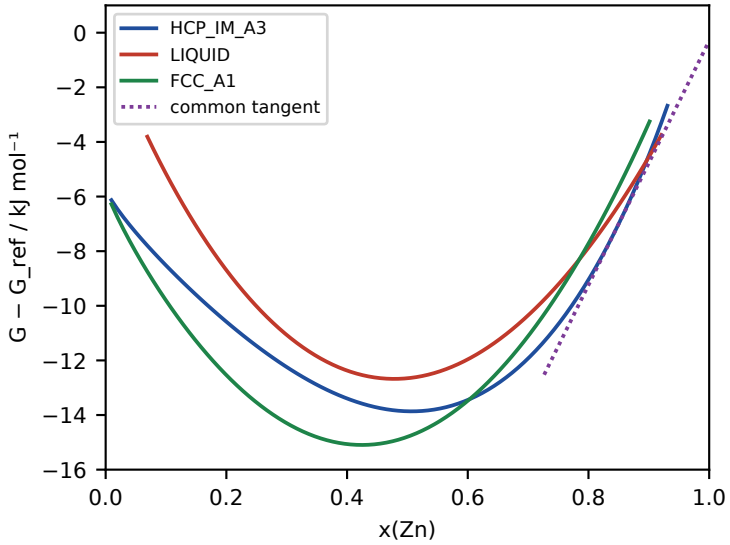


Fig. 4. Calculated Gibbs energy curves at 760 K.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Cu–Zn).

Phase	at.% Zn	ΔfH(298.15 K) / kJ mol⁻¹	ΔfS / J mol⁻¹ K⁻¹	Cp(298.15 K) / J mol⁻¹ K⁻¹
β	49.7	-9.43	4.39	24.92
δ	75.0	-5.76	5.22	25.16
γ	64.0	-10.64	2.56	25.05
β′	46.0	-10.41	2.74	24.88
ε	83.0	-4.89	3.42	25.24

Reference: stable elements (SER) at 298.15 K; values per mole of atoms.

Table 4. Integral and partial mixing properties of the liquid at 1450 K (Cu–Zn).

	Liquid at 1450 K					
at.% Zn	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Cu)	a(Zn)	ΔH(Cu) / kJ mol⁻¹	ΔH(Zn) / kJ mol⁻¹
10	-2.62	-6.10	0.8862	0.0189	-0.33	-23.20
20	-4.56	-9.98	0.7446	0.0518	-1.37	-17.35
30	-5.82	-12.60	0.5856	0.1069	-3.09	-12.17
40	-6.39	-14.11	0.4278	0.1916	-5.41	-7.86
50	-6.32	-14.55	0.2894	0.3091	-8.13	-4.52
60	-5.69	-13.96	0.1810	0.4539	-10.98	-2.17
70	-4.60	-12.34	0.1041	0.6112	-13.59	-0.75
80	-3.17	-9.68	0.0536	0.7621	-15.50	-0.09
90	-1.58	-5.87	0.0215	0.8923	-16.17	0.04

Reference states: pure liquid Cu and Zn at 1450 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] W. Gierlotka, S.-W. Chen, J. Mater. Res., 23, 258-63(2008).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.